

UNISOLE AI CAMPUS PROGRAM

GROUP 1 — COMPUTER SCIENCE & IT

Pathway 2: Full Stack Web Development

Target students: BCA • MCA • B.Sc. CS/IT • PGDCA • B.Tech CSE/IT • Related disciplines

Format: Weekday technical track (can be paired with the weekend AI Entrepreneurship & Innovation track)

Overview

This pathway is for students who want to build complete, production-style web products — the kind of applications most companies actually hire developers to build and maintain. It moves from front-end fundamentals to backend systems to databases, then layers in AI features so graduates leave with modern, AI-integrated web development skills rather than a purely legacy stack.

Why this pathway matters: Full stack developers remain one of the most consistently hired technical roles, but the bar has moved — companies now expect developers who can also integrate AI features into their products. This pathway closes that gap directly.

Module 1 — Web Foundations

The building blocks of every website: how a browser renders content, how pages respond to user interaction, and how a page talks to a server.

Topics covered:

- HTML and Semantic HTML — structuring a web page meaningfully, not just visually
- CSS and Responsive design — styling pages that work across screen sizes
- JavaScript fundamentals — the programming language of the browser
- DOM and Events — how JavaScript interacts with a live page
- Async JavaScript — handling operations that take time, like network requests
- Fetch/API consumption — getting data from a server into a page

Skills gained: The ability to build a functional, responsive web page from scratch and understand exactly what happens when a browser loads it.

Module 2 — Modern Frontend

Students learn React, the framework used by most modern web companies, to build interactive, component-based user interfaces instead of static pages.

Topics covered:

- React — the industry-standard library for building user interfaces
- Components, Props, State, Hooks — the core building blocks of a React application

- Forms — capturing and validating user input
- Routing — navigating between pages in a single-page application
- API integration — connecting the frontend to real backend data
- Vite — a modern, fast frontend build tool
- Frontend project structure — organizing a real-world React codebase

Skills gained: The ability to build a modern, interactive user interface the way professional frontend teams do.

Module 3 — Backend Development

The server side of the application — how requests are received, processed, secured, and answered.

Topics covered:

- Node.js — running JavaScript on the server
- Express — the most widely used Node.js web framework
- REST APIs — the standard architecture for web services
- Middleware — code that runs between a request and a response
- Authentication and Authorization — verifying who a user is and what they're allowed to do
- Validation — rejecting bad input before it causes problems
- Error handling — responding safely and clearly when something goes wrong
- API documentation — making an API usable by other developers

Skills gained: The ability to design and build a secure, well-structured backend service from scratch.

Module 4 — Databases

Every application needs a place to persist data. This module covers MongoDB, a widely used NoSQL database, and how to design and query it correctly.

Topics covered:

- Database fundamentals — how data is stored and retrieved
- MongoDB — a flexible, document-based database used widely in modern web stacks
- CRUD — Create, Read, Update, Delete, the four core data operations
- Schema design — structuring data for both performance and clarity
- Indexing — making queries fast at scale
- Aggregation — summarizing and transforming data inside the database itself
- Connecting backend with database — wiring the whole data layer together

Skills gained: The ability to design a database schema and connect it reliably to a real application.

Module 5 — Software Engineering

The professional practices that separate a hobby project from a maintainable, team-ready codebase.

Topics covered:

- Git, GitHub, Branching, Pull requests — how real teams collaborate on code

- Code organization — structuring a project so it stays manageable as it grows
- Environment variables — managing configuration and secrets safely
- Debugging — systematically finding and fixing problems
- Testing fundamentals — verifying code actually works before it ships

Skills gained: The engineering discipline expected of a junior developer on a real team, not just someone who can make code run once.

Module 6 — Deployment

Turning a local project into a live, publicly accessible application.

Topics covered:

- Production builds — preparing an application to run outside a development environment
- Hosting — getting an application onto the internet
- Environment configuration — managing settings across development and production
- Docker fundamentals — packaging an application so it runs consistently anywhere
- CI/CD concepts — automating testing and deployment
- Monitoring basics — knowing when a live application breaks

Skills gained: The ability to take a finished project and actually ship it — a step many student projects never reach.

Module 7 — AI-Powered Web Applications

The differentiator: students add real AI capability into their web app rather than treating it as a separate skill.

Topics covered:

- Connecting LLM APIs — calling large language models from a web application
- AI features inside web applications — practical integration patterns
- RAG-based applications — building AI features grounded in real documents
- AI assistants and Document Q&A; — common, high-value AI product patterns
- AI-powered dashboards — surfacing AI-generated insight inside a real interface

Skills gained: The ability to add genuinely useful AI functionality to a web product — a skill that increasingly separates competitive developer candidates from the rest.

Capstone Project

Build a complete: **Frontend + Backend + API + Database + AI Feature + Deployment**

This capstone requires students to own every layer of a real application — from the interface a user sees, to the server logic behind it, to the database storing its data, to at least one genuine AI-powered feature, all the way through to a live, deployed product.

Tools & technologies used across this pathway: HTML, CSS, JavaScript, React, Vite, Node.js, Express, MongoDB, Git/GitHub, Docker, LLM APIs.

Career relevance: Full Stack Developer, Frontend Developer, Backend Developer, Web Application Developer, AI-Integrated Product Developer.

Industry-Readiness Layer (*applies alongside this pathway*)

Professional Digital Profile: Professional email, LinkedIn, Resume, Portfolio, Project documentation, Professional communication.

GitHub / Evidence: Git, GitHub, README, Repository structure, Version control.

Interview Preparation: Resume walkthrough, Tell-me-about-yourself, Project explanation, Behavioural questions, Technical/domain questions, Mock interview, Feedback.

Industry Interaction: Weekly/periodic expert sessions, Career sessions, Industry case studies, How companies recruit, How to approach professionals, Networking etiquette.

Project Defence: Every student must be able to answer — What problem did you solve? Why did you choose this approach? What did you build? What did you learn? What are the limitations? What would you improve?

Final Capstone Framework

1. **Problem** — Identify a real-world problem
2. **Research** — Understand the domain and existing solutions
3. **Data/Information** — Collect and organize relevant information
4. **Build** — Create the appropriate technical solution
5. **Validate** — Test the solution and interpret results
6. **Document** — Report, Repository/portfolio, Documentation, Presentation
7. **Defend** — Present the project before an evaluator/mentor

What This Student Should Have at the End

GitHub + Software Project + AI/ML Project + Deployment where applicable + Resume

Training and assessment are program components. Internship, advanced projects, hackathons, research engagement and other opportunities are performance/availability-based and subject to applicable selection procedures.